Problem:

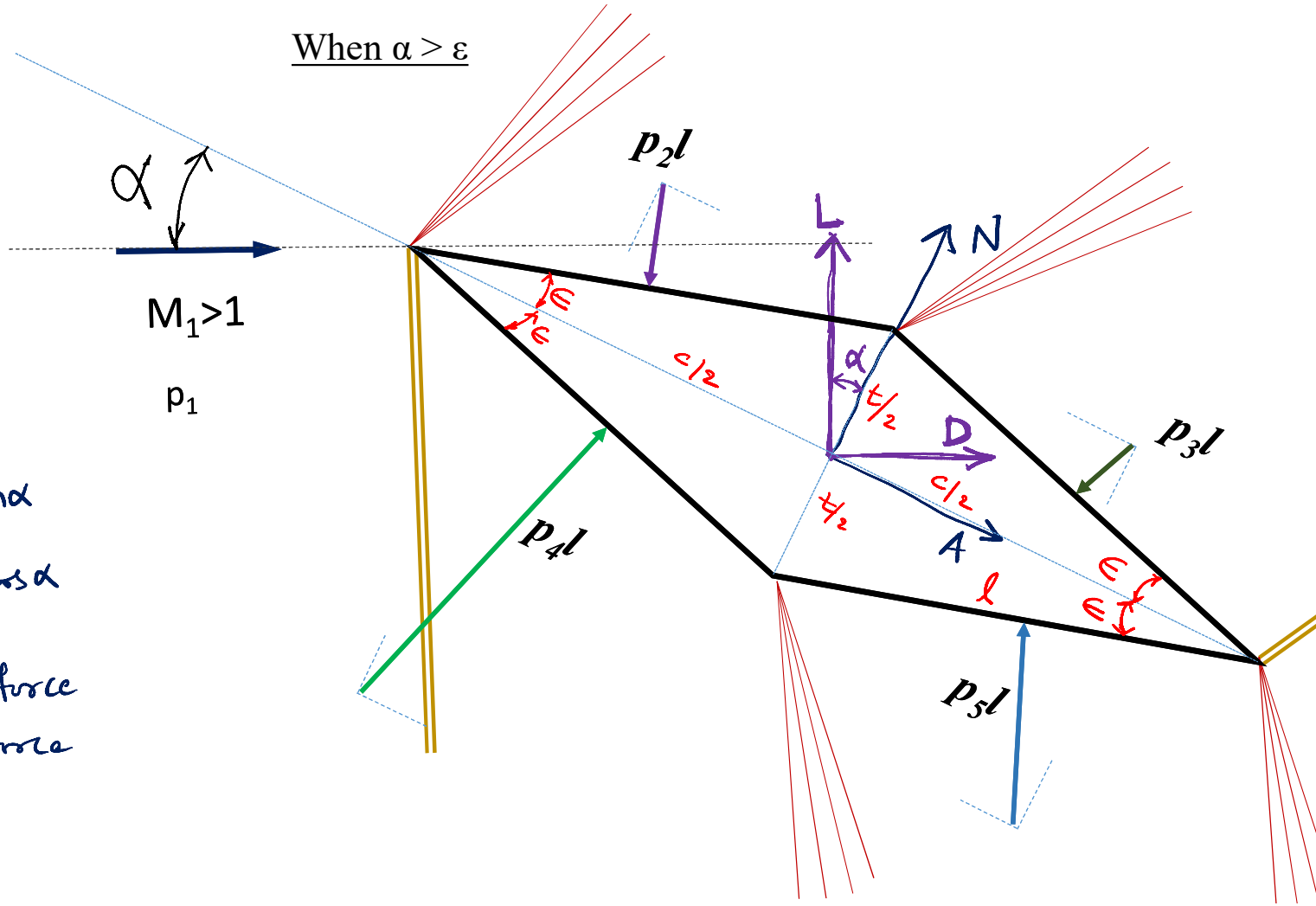
A diamond airfoil is at angle of attack of 15 deg in Mach 3 flow. The half angle of the diamond airfoil is 10 deg. Calculate the Lift and Drag coefficients.

$M_1 = 3$

$\alpha = 15^\circ$

$\epsilon = 10^\circ$

$\alpha > \epsilon$

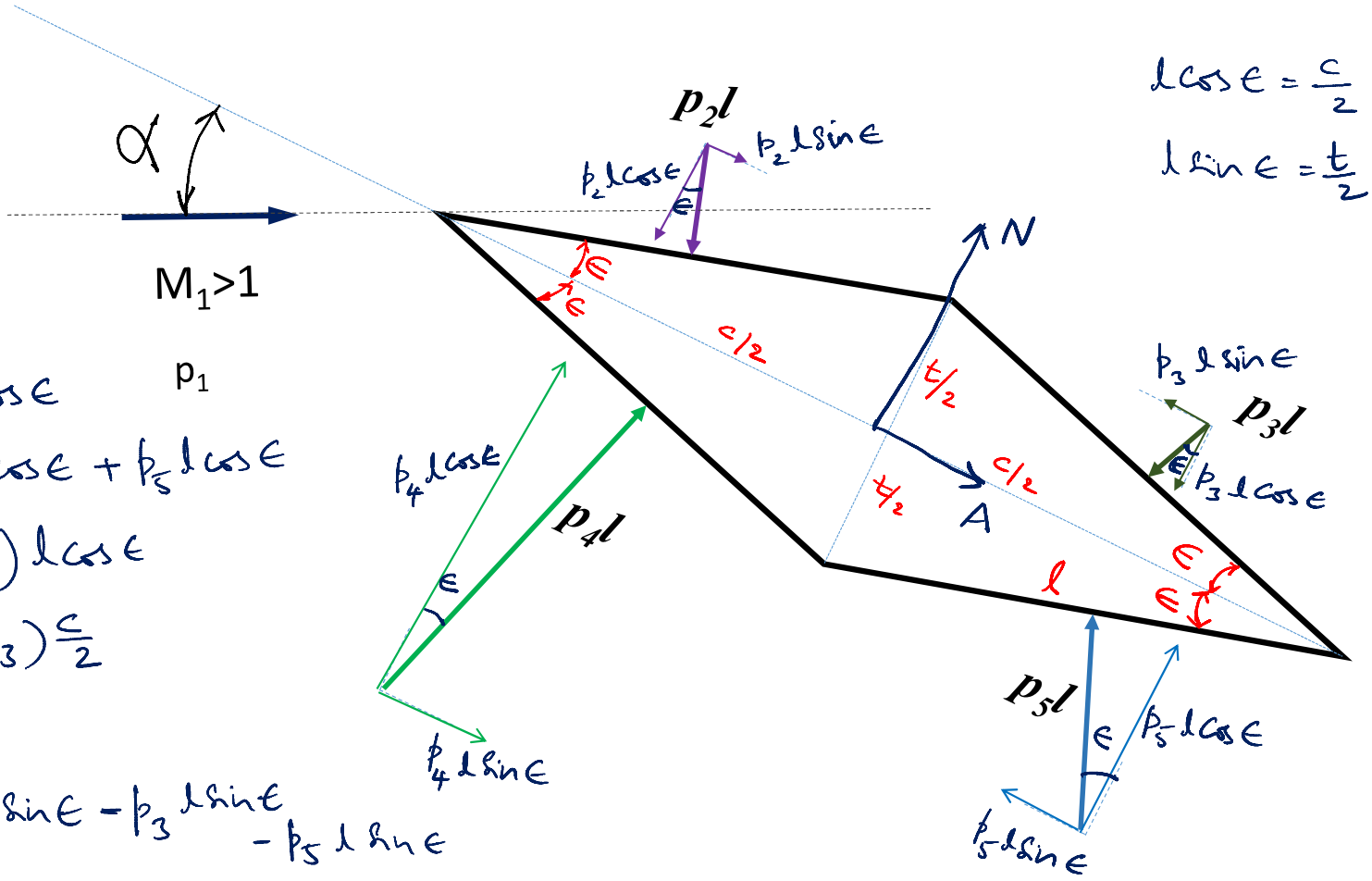


Lift $L = N\cos\alpha - A\sin\alpha$

Drag $D = N\sin\alpha + A\cos\alpha$

Where N – Normal force

A – Axial force



$l\cos\epsilon = \frac{c}{2}$

$l\sin\epsilon = \frac{t}{2}$

Normal force N:

$$N = -p_2 l\cos\epsilon - p_3 l\cos\epsilon + p_4 l\cos\epsilon + p_5 l\cos\epsilon$$

$$N = (p_4 + p_5 - p_2 - p_3) l\cos\epsilon$$

$$N = (p_4 + p_5 - p_2 - p_3)\frac{c}{2}$$

Axial force A:

$$A = p_2 l\sin\epsilon + p_4 l\sin\epsilon - p_3 l\sin\epsilon - p_5 l\sin\epsilon$$

$$A = (p_2 + p_4 - p_3 - p_5) l\sin\epsilon$$

$$A = (p_2 + p_4 - p_3 - p_5)\frac{t}{2}$$

Lift Coefficient

$$C_l = \frac{L}{q_1 c} = \frac{N\cos\alpha - A\sin\alpha}{\frac{1}{2}\gamma p_1 M_1^2 c}$$

$$\left[q_1 = \frac{1}{2}\rho_1 V_1^2 = \frac{\gamma_1 p_1 M_1^2}{2} \right]$$

$$= \frac{2}{\gamma M_1^2 c}\left[\left(\frac{p_4}{p_1} + \frac{p_5}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1}\right)\frac{c}{2}\cos\alpha - \left(\frac{p_2}{p_1} + \frac{p_4}{p_1} - \frac{p_3}{p_1} - \frac{p_5}{p_1}\right)\frac{t}{2}\sin\alpha\right]$$

$$\Rightarrow C_l = \frac{1}{\gamma M_1^2}\left[\left(\frac{p_4}{p_1} + \frac{p_5}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1}\right)\cos\alpha - \left(\frac{p_2}{p_1} + \frac{p_4}{p_1} - \frac{p_3}{p_1} - \frac{p_5}{p_1}\right)\frac{t}{c}\sin\alpha\right] \quad -(1)$$

Drag Coefficient

$$C_d = \frac{D}{q_1 c} = \frac{N\sin\alpha + A\cos\alpha}{\frac{1}{2}\gamma p_1 M_1^2 c}$$

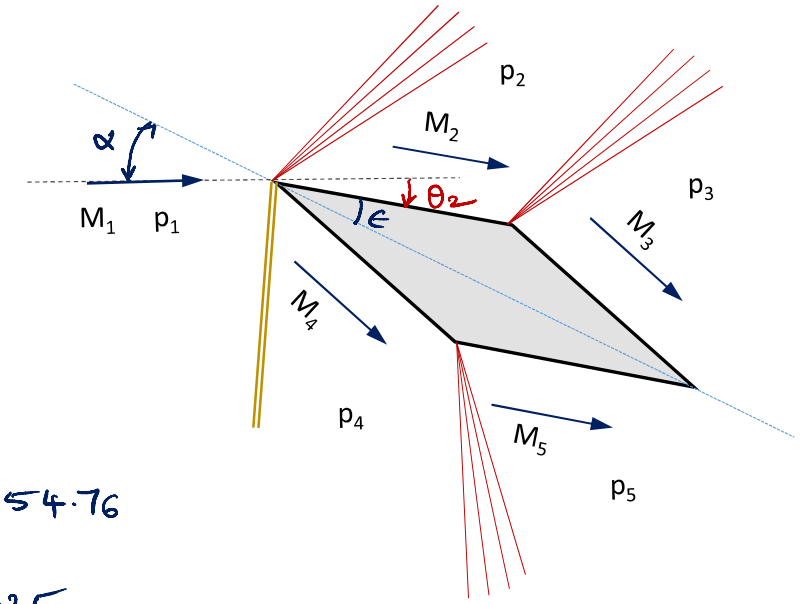
$$\Rightarrow C_d = \frac{1}{\gamma M_1^2}\left[\left(\frac{p_4}{p_1} + \frac{p_5}{p_1} - \frac{p_2}{p_1} - \frac{p_3}{p_1}\right)\sin\alpha + \left(\frac{p_2}{p_1} + \frac{p_4}{p_1} - \frac{p_3}{p_1} - \frac{p_5}{p_1}\right)\frac{t}{c}\cos\alpha\right] \quad -(2)$$

$\frac{p_2}{p_1}$:

1–2 : Expansion fan

Flow deflection angle $\theta_2 = \alpha - \epsilon = 15 - 10$

$= 5°$



- For $M_1 = 3$ $\xrightarrow{\text{P–M tables}}$ $\nu(M_1) = 49.76$
- $\theta_2 = \nu(M_2) - \nu(M_1) \Rightarrow \nu(M_2) = \theta_2 + \nu(M_1)$

  $= 5 + 49.76 = 54.76$
- For $\nu(M_2) = 54.76$ $\xrightarrow{\text{P–M tables}}$ $M_2 = 3.25$
- For $M_1 = 3$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_1}{p_{01}} = 0.0272$

  For $M_2 = 3.25$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_2}{p_{02}} = 0.0188$

$\Rightarrow \frac{p_2}{p_1} = \frac{p_2/p_{02}}{p_1/p_{01}}$ $(\because p_{01} = p_{02})$

$= \frac{0.0188}{0.0272} = 0.69$

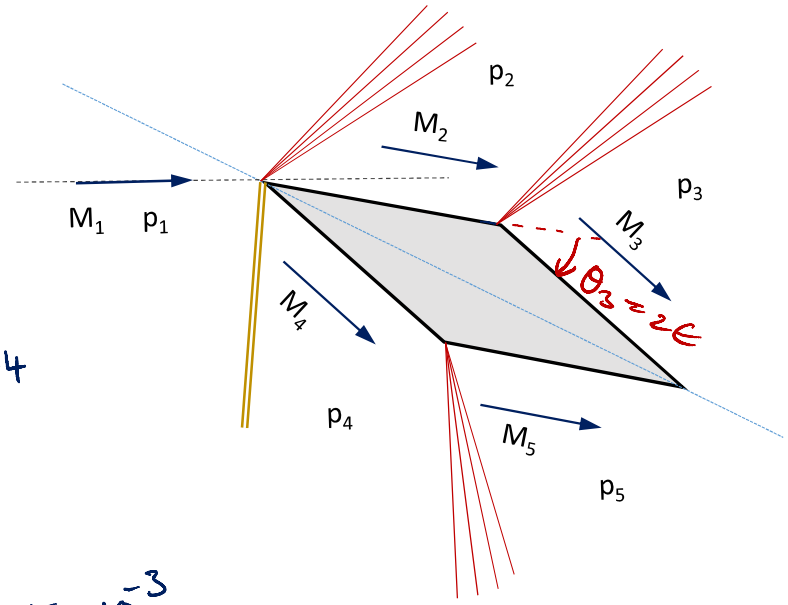
$\Rightarrow \boxed{\frac{p_2}{p_1} = 0.69}$ — (3)

$\frac{p_3}{p_1}$:

2-3 : Expansion fan

$\theta_3 = 2\epsilon = 20^\circ$

- For $M_2 = 3.25$ $\xrightarrow{\text{P-M Tables}}$ $\nu(M_2) = 54.4$
- $\theta_3 = \nu(M_3) - \nu(M_2) \Rightarrow \nu(M_3) = \theta_3 + \nu(M_2)$
  $= 20 + 54.4 = 74.4$
- For $\nu(M_3) = 74.4$ $\xrightarrow{\text{P-M Tables}}$ $M_3 = 4.75$
- For $M_3 = 4.75$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_3}{p_{03}} = 2.675 \times 10^{-3}$



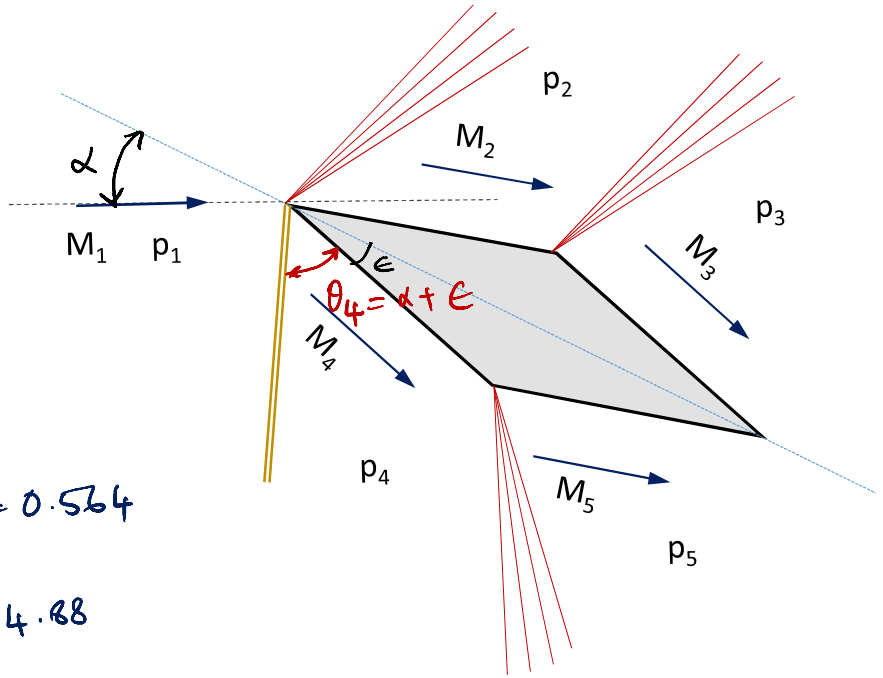
$$\frac{p_3}{p_1} = \frac{p_3/p_{03}}{p_1/p_{01}} \quad (\text{since } p_{01} = p_{02} = p_{03})$$

$$= \frac{2.675 \times 10^{-3}}{0.0272} = 0.0983$$

$$\boxed{\frac{p_3}{p_1} = 0.0983} \quad (4)$$

$\frac{p_4}{p_1}$: 1 – 4 : oblique shock

$\theta_4 = \alpha + \epsilon = 15 + 10 = 25^\circ$



- For $M_1 = 3$ and $\theta_4 = 25^\circ$ $\xrightarrow{\theta\text{-}\beta\text{-}M \text{ Chart}}$ $\beta_4 = 44^\circ$
  - $M_{n_1} = M_1 \sin\beta_4 = 3 \sin 44 = 2.08$
  - For $M_{n_1} = 2.08$ $\xrightarrow{\text{Normal Shock Tables}}$ $M_{n_4} = 0.564$

$$\frac{p_4}{p_1} = 4.88$$

$$\frac{p_{04}}{p_{01}} = 0.684$$

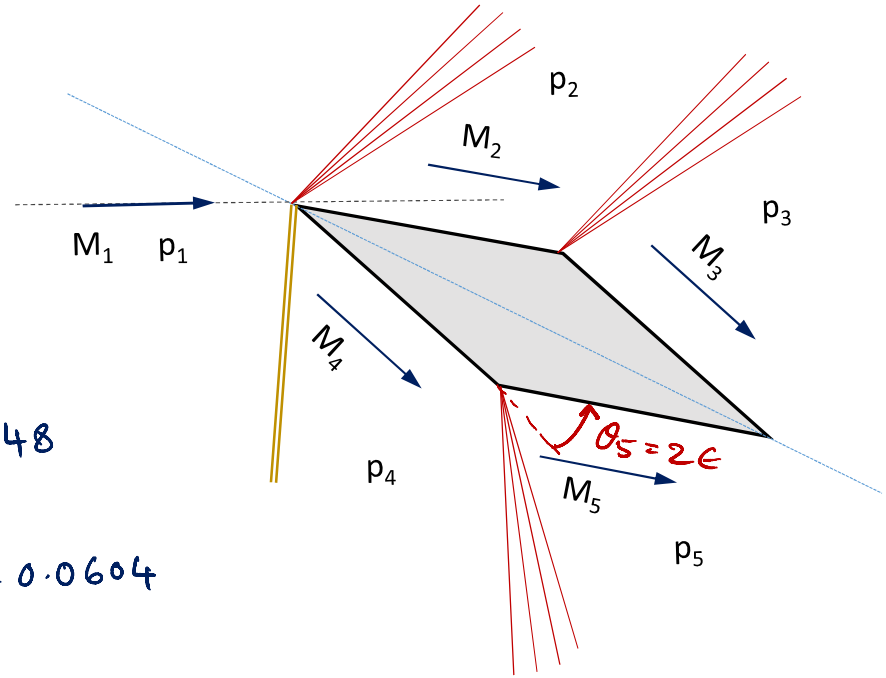
- $M_4 = \dfrac{M_{n_4}}{\sin(\beta_4 - \theta_4)} = \dfrac{0.564}{\sin(44 - 25)} = 1.733$

$$\boxed{\frac{p_4}{p_1} = 4.88} \quad (5)$$

$\frac{p_5}{p_1}$: 4-5 Expansion fan

$\theta_5 = 2\epsilon = 20^\circ$

- For $M_4 = 1.73$ $\xrightarrow{\text{P-M Tables}}$ $\nu(M_4) = 18.7$
- $\nu(M_5) = \theta_5 + \nu(M_4) = 20 + 18.7 = 38.7$
- For $\nu(M_5) = 38.7$ $\xrightarrow{\text{P-M Tables}}$ $M_5 = 2.48$
- For $M_5 = 2.5$ $\xrightarrow{\text{Isentropic tables}}$ $\frac{p_5}{p_{05}} = 0.0604$



$$\frac{p_5}{p_1} = \frac{p_5}{p_{05}} \times \frac{p_{04}}{p_{01}} \times \frac{p_{01}}{p_1} \quad (\text{Since } p_{04} = p_{05})$$

$$= 0.0604 \times 0.684 \times \frac{1}{0.0272} = 1.5$$

$$\boxed{\frac{p_5}{p_1} = 1.5} \quad \text{—— (6)}$$

$\gamma = 1.4$ $\quad \frac{p_2}{p_1} = 0.69$ $\quad \frac{p_3}{p_1} = 0.0983$ $\quad \frac{t}{c} = \tan\epsilon = \tan 10^\circ$

$M_1 = 3$

$\alpha = 15^\circ$ $\quad \frac{p_4}{p_1} = 4.88$ $\quad \frac{p_5}{p_1} = 1.5$

Substituting all the above values in equations ① and ② yields C_l and C_d as

$C_l = 0.414$

$C_d = 0.169$